

Superconducting Whispering Gallery Mode Resonators

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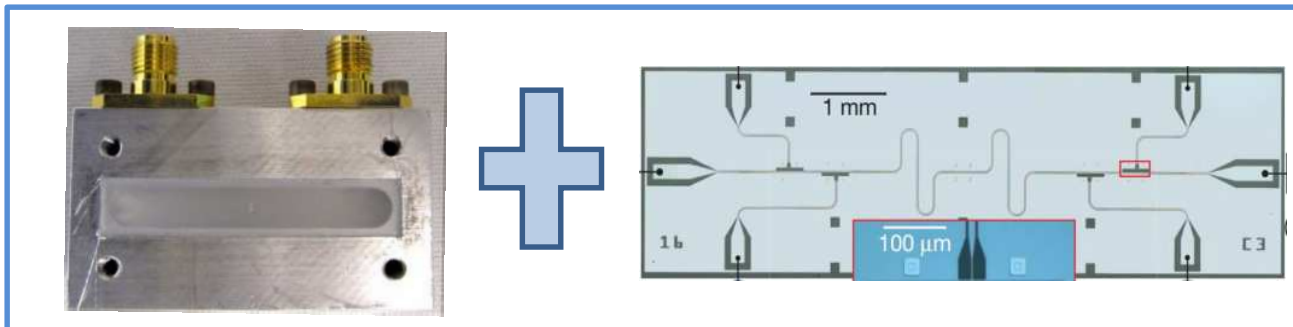
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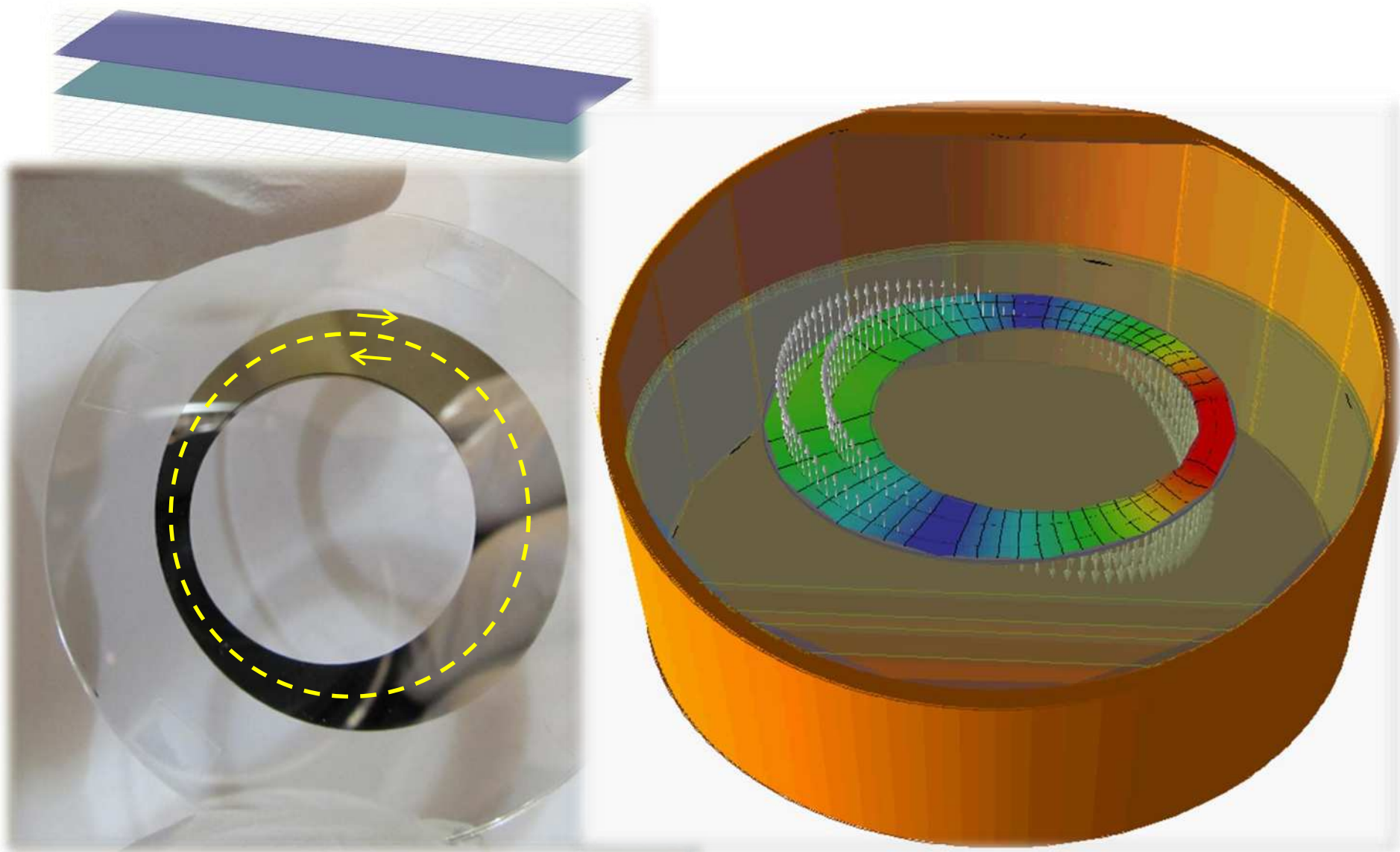
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Motivation

- I. Marry wafer scalability & 3D quality?
 - Coupled resonators & large number of qubits
 - Flux bias lines
- II. Test materials at the single photon level
 - Thin films
 - Dielectrics

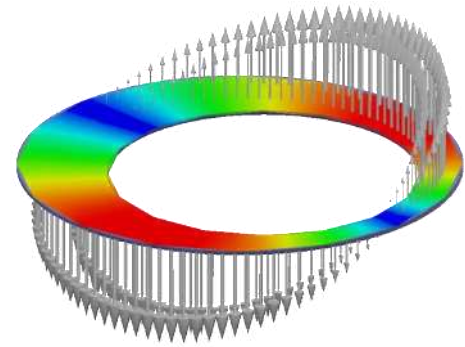


Towards 'Wafer-Scalable' cQED architecture



There are **2 rings** in this picture!

Differential Mode



Sample Holder

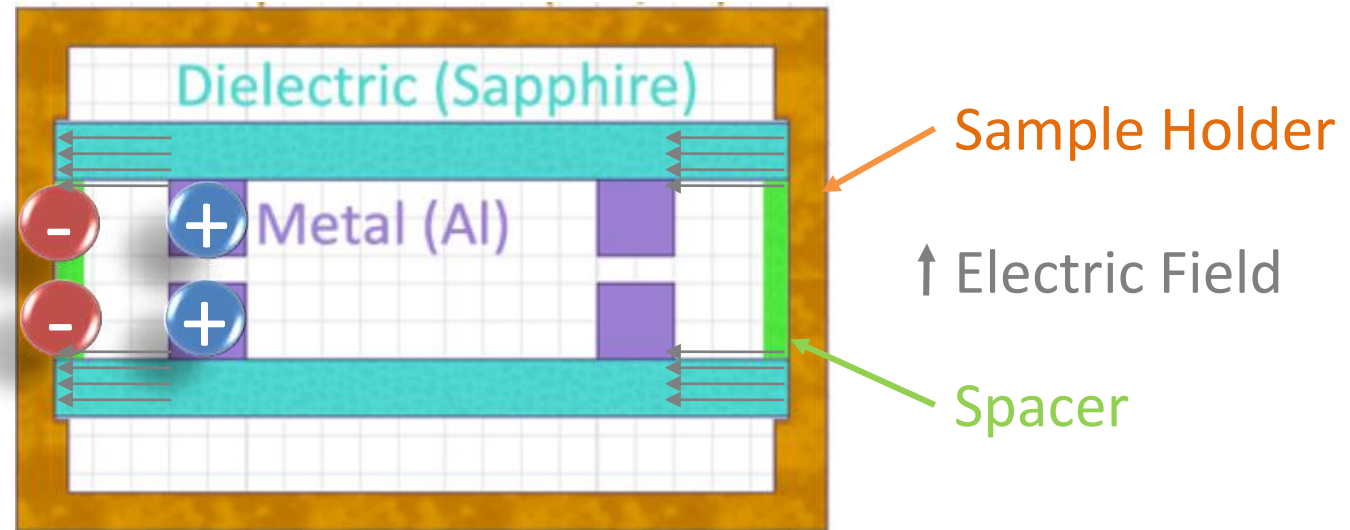
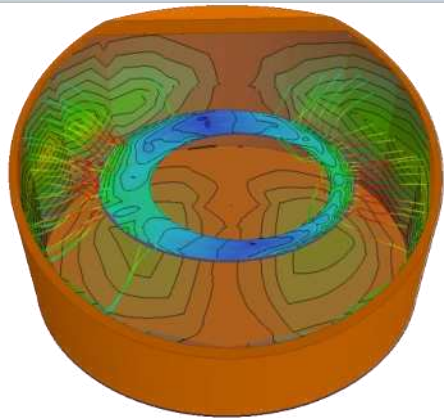
↑ Electric Field

Spacer

Not To Scale!

- 98 % of field energy is in vacuum

Common Mode



Not To Scale!

- 44% of field energy is in dielectric.

Modular Cavity & Experimental Sample Holder



**Non Magnetic
SMA Pin Coupler**

Technical Parameters:

Al film thickness	300 nm
Sapphire wafer thickness	450 μm
Wafer spacing	200 μm
Wafer diameter	2.0 in

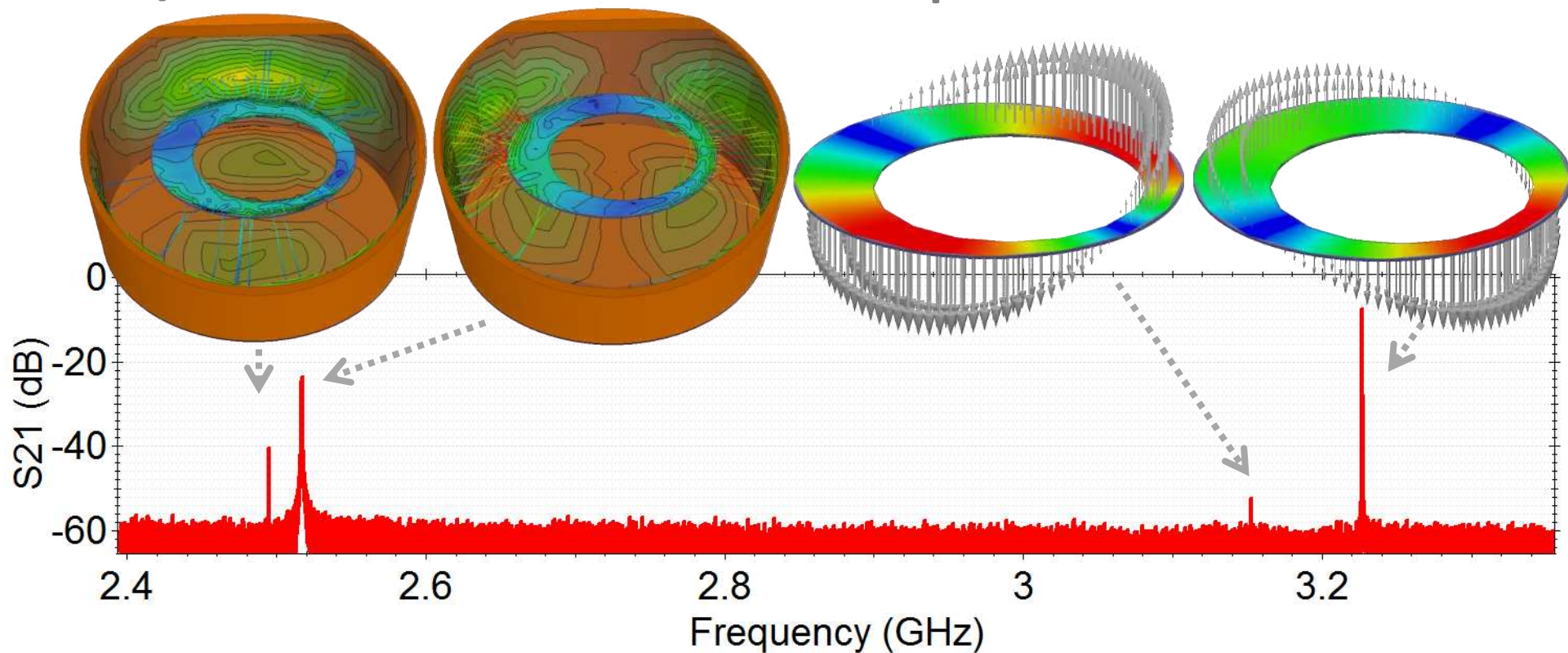
Identifying Modes (S_{21})

Common
Perpendicular

Common
Parallel

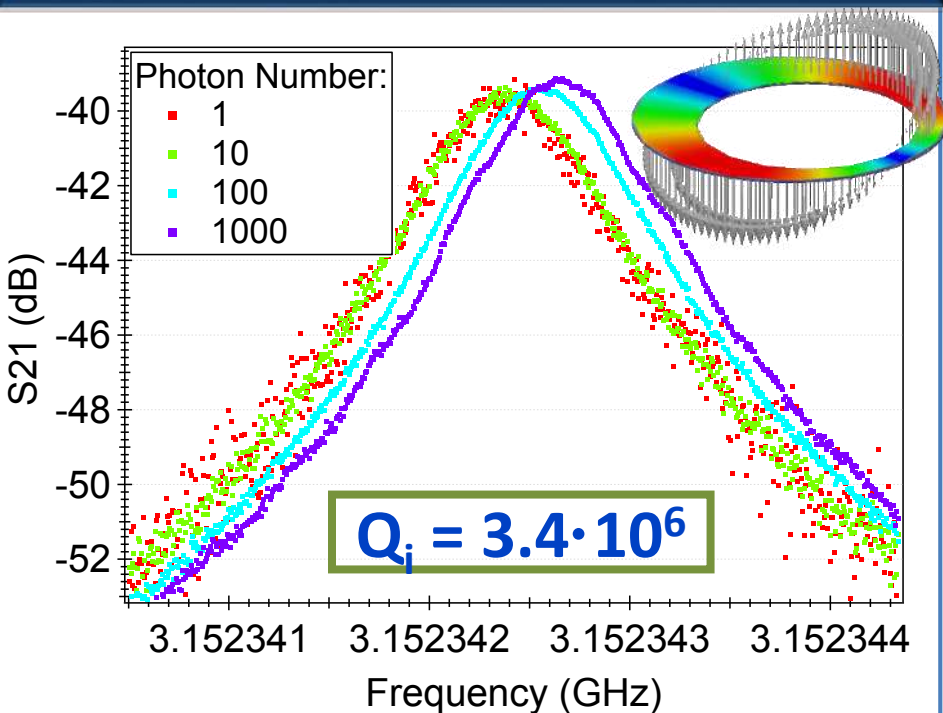
Differential
Perpendicular

Differential
Parallel



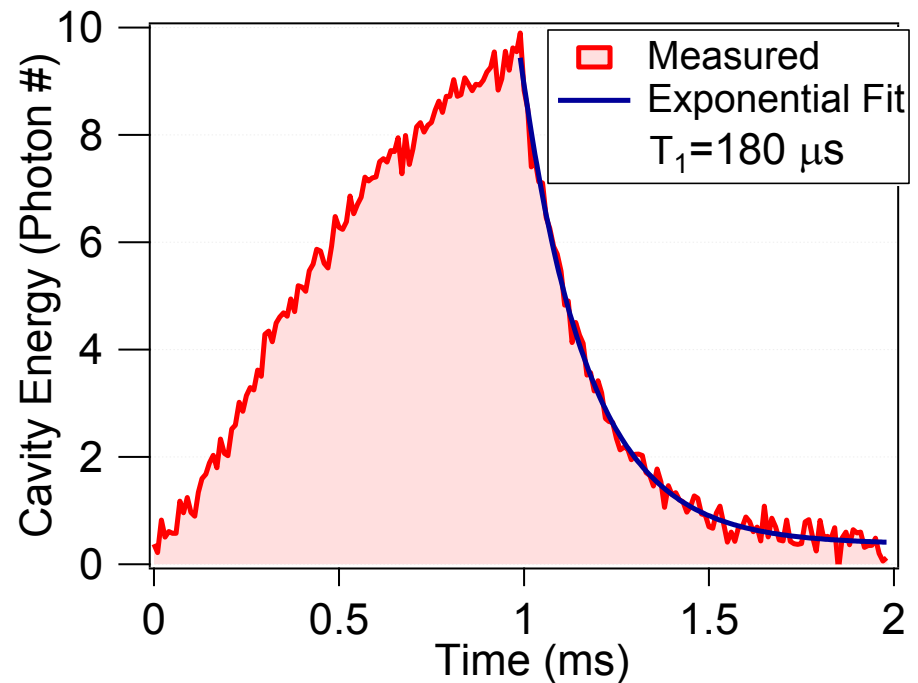
- Differential modes contain > 98% of mode energy in between the rings

Total Q vs Power (S_{21})

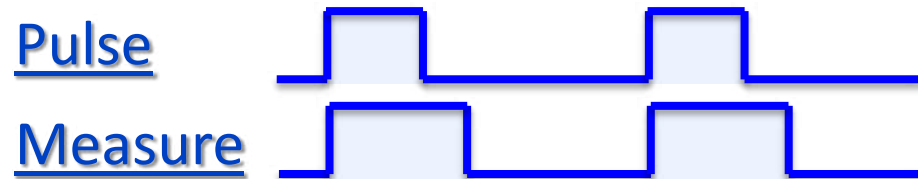


- Undercoupled differential perpendicular mode
- Aluminum sample holder

Energy Relaxation (T_1)

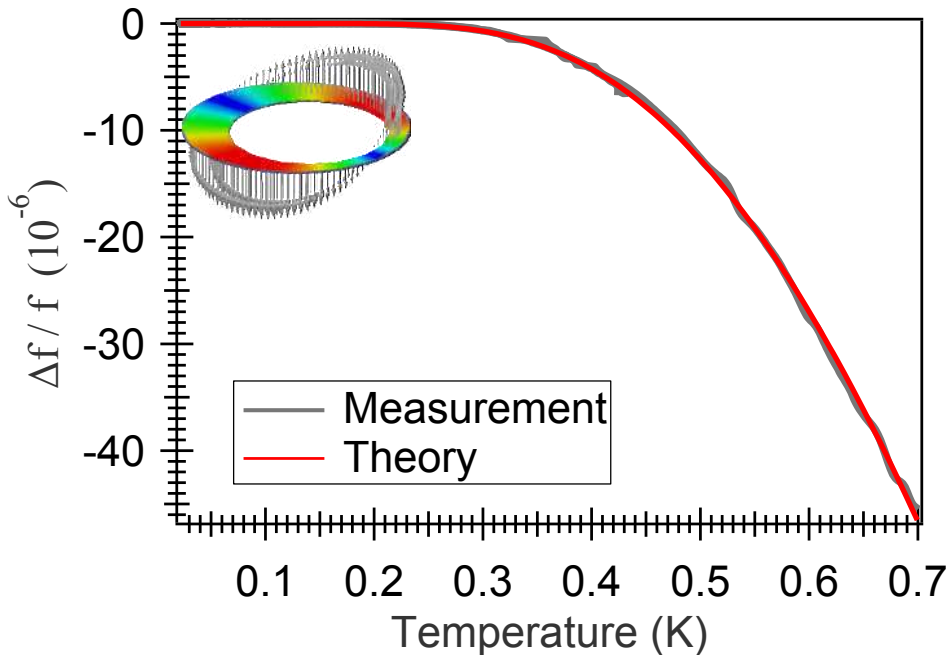


- Heterodyne Measurement
- $T_2 \cong 2 T_1$



Aluminum

Differential perpendicular mode

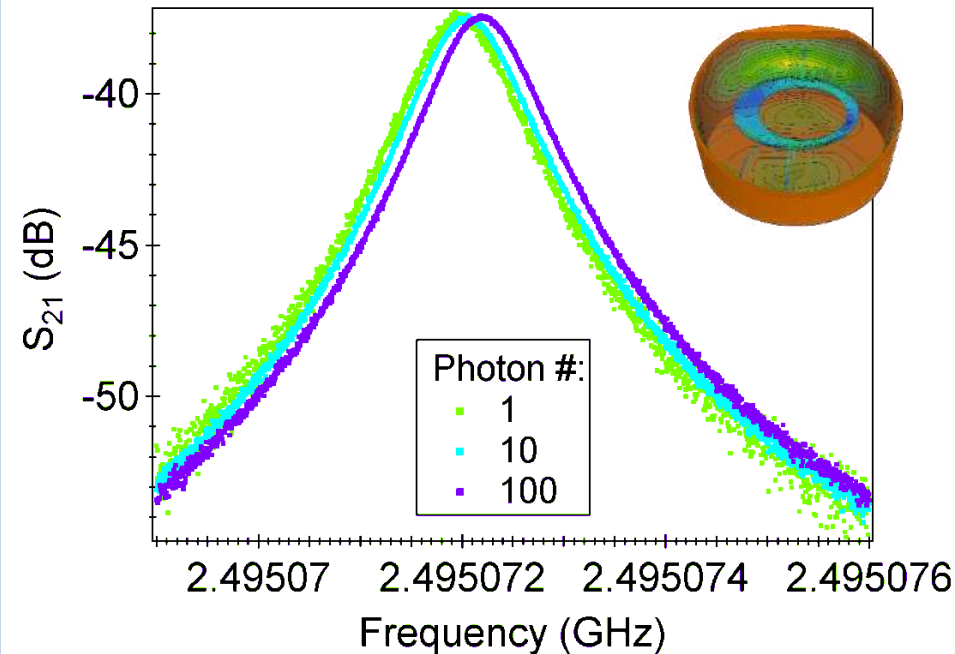


- Extracted upper bound of the surface square resistance of thin film aluminum:

$$R_{\square} \leq 250 \text{ n}\Omega$$
$$Q_s > 5000$$

Sapphire

Common perpendicular mode



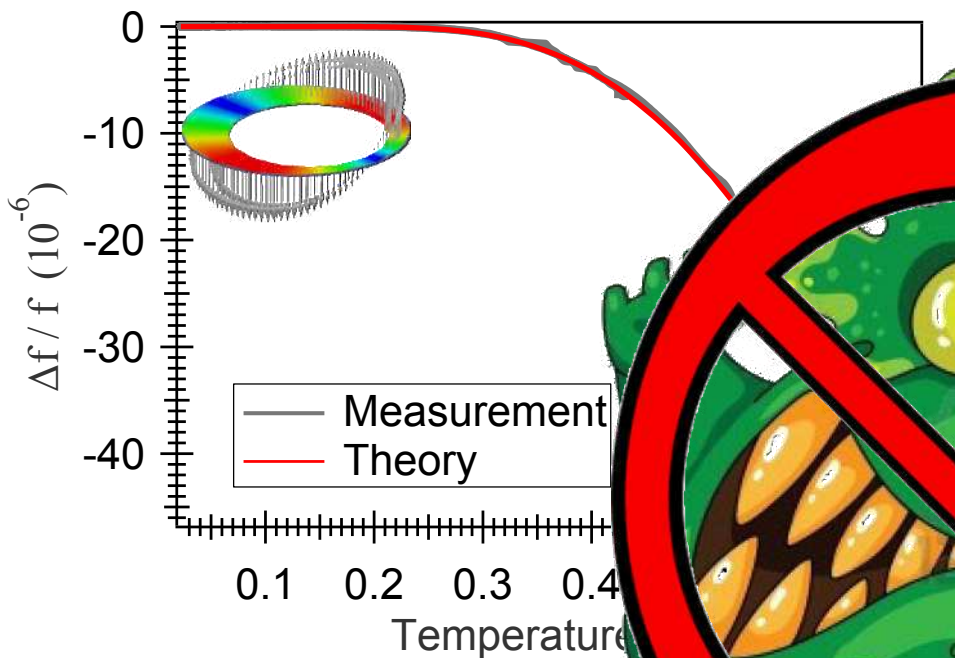
- HFSS participation ratio: 44%
- Extracted sapphire properties at the single photon level:

$$Q = 2.1 \cdot 10^6$$
$$\tan \delta < 1 \cdot 10^{-6}$$

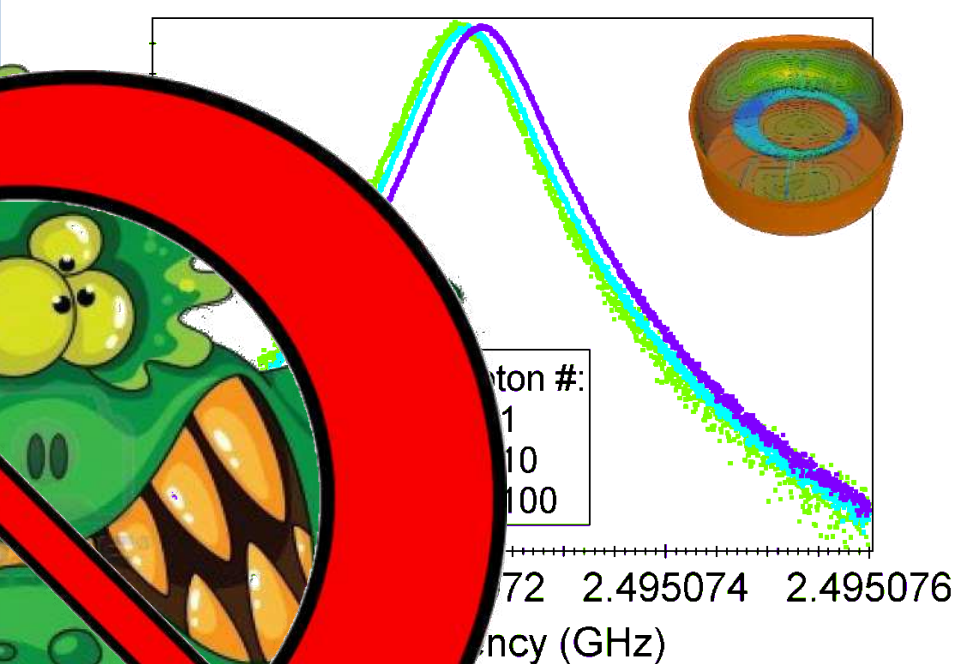
Aluminum

Sapphire

Differential perpendicular mode



Common perpendicular mode



- Extracted upper bound for surface square resistance of film aluminum: $1.5 \times 10^{-4} \Omega/\square$ at photon level: 10^{18} cm^{-2} (approximate sapphire properties at photon level: 10^{18} cm^{-2})

$$R_{\square} \leq 250 \text{ n}\Omega$$
$$Q_s > 5000$$

$$Q = 2.1 \cdot 10^6$$
$$\tan \delta < 1 \cdot 10^{-6}$$

Future Directions of `2.5D` Resonators

- Integration with multiple qubits and resonators
- Quantum Bus
- Study of *cryogenic material properties*
- *Galvanic cavity-qubit coupling*

